05. Train-Test Split & Validation Strategy

Understand the 70/15/15 split and why stratification matters.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Split implemented

The main and SMOTE notebooks use train_test_split with test_size=0.30, random_state=42, followed by splitting the temporary 30% equally into validation and test. For 837 rows, the recorded partitions are 585 training, 126 validation and 126 test observations.

Class counts in the SMOTE notebook

Class	Train before SMOTE	Validation	Test
Overweight	66	10	18
Stunting	485	110	101
Underweight	31	5	7
Wasting	3	1	0
Total	585	126	126

Critical consequence

The test partition contains zero Wasting examples. No test-set recall, precision or F1 for Wasting can be estimated from this split. The validation partition has only one Wasting example, so a single prediction changes its recall from 0 to 1. This is not a reliable estimate of rare-class generalisation.

Recommended split design

- Use stratify=y in both train_test_split calls so class proportions are approximately preserved, while recognising that four Wasting records are still too few for robust evaluation.
- If multiple rows can belong to one child, split by child/group to prevent identity leakage.
- Consider repeated stratified cross-validation for model selection, but report the instability caused by the four-record class explicitly.
- Acquire more rare-class examples before claiming performance on Wasting.
- Keep a final untouched test set; never choose hyperparameters based on it.

Interview response

> “The notebook uses a 70/15/15 split with random_state 42. The important issue is that it does not stratify, and the test partition ends up with no Wasting examples. Therefore, I cannot claim the model generalises to Wasting based on that test score. I would redesign the evaluation with stratification and, more importantly, collect substantially more rare-class data.”